Fieldwork maintenance and procedures

This document summarizes the maintenance procedures needed in each site. Please consult the detail versions for specific site, and sensor's maintenance and diagnostics.

After conducting any maintenance protocol, register it in the SharePoint field notes to have a follow-up of each sensor and station. If any equipment issues arise, please contact David Trejo Cancino (trejo_cancino.david_ulises@uqam.ca) to log the problem in the database and proceed with the required diagnostics and repairs.

Before performing any maintenance please thoroughly check the details of that procedure in this document and the detailed version. If any questions arise, do not hesitate to contact David Trejo (trejo_cancino.david_ulises@uqam.ca).

Data from all sites are currently transmitted remotely to our servers. Field data downloads should be performed as a backup and to clear memory, ensuring SmartFlux operates efficiently without lag.

Routine Maintenance and Procedures Across All Sites

Standard maintenance visits are recommended every 4 to 6 weeks at each site, with additional trips scheduled as necessary. Prior to fieldwork, carefully review all key procedural information. Use the provided checklists to prepare for site visits and to ensure consistent application of procedures.

Checklists

Use the following checklists to help plan and conduct site visits.

Pre-Departure

This table is best completed the afternoon/evening before a site visit. Although you may not be able to pack the truck until the morning depending on scheduling, you can still gather supplies.

Item	Details
Field Laptop	Charge it overnight and bring cables plus an external charger.
USB Sticks	2 per site, ensure they are empty and formatted in fat32.
Cleaning supplies	Microfiber cloths + washer fluid as needed.
Desiccant	4-5 bags per sites every two visits.
First Aid Kit	
Waders / Boots / PPE	Take PPE according to the site and the tasks that are going to be performed on site.
Spare socks	In every visit

Torch or de-icing fluids for locks	In winter
Tool kit	Check if any specific tools are needed for the intended tasks.
Fire extinguisher and water spray	During fire season
Other items	Check if there is any item needed for the tasks planned onsite.

On-site

Task	Details
Clean IRGAs as needed	Follow the detailed instructions in the corresponding documents.
Calibrate IRGAs	Follow the detailed instructions in the corresponding documents.
Swap USB sticks in 7550	Press eject and wait for light to turn off before removal. Ensure light is blinking after inserting the new stick.
Refill 7700 washer fluid	Take note of fluid supply on site.
Check radiometers	Clean as needed.
Check rain gauge	Clean as needed.
Tight guywires	Tense the turnbuckles as needed.
Check Battery Box	Check correct functioning and that there are no other issues.
Note anything unusual	Take notes of potential improvements and issues

After trip

Task	Details
Update the team	Text in Teams to team and share important updates.
Fill out field logs	Fill field notes in SharePoint
Return vehicle	Return the vehicle and follow the protocol on returning it.
Transfer data	Transfer data to the local copy
Charge field laptop	
Clean and sort tools	As needed.

Cleaning and maintenance procedures

- Clean the methane sensor LI-7700 in every visit following the instructions detailed in the “Maintenance_LI7700” document.
- Clean the LI-7200RS gas analyzer when needed following the instructions detailed in the “Maintenance_LI7200RS” document. After cleaning the LI-7200RS a calibration of the sensor is mandatory.

- Clean the CNR4 net radiometer every two visits or when necessary, following the instructions detailed in the “Maintenance_biomet_sensors” document.
- Follow the maintenance procedures detailed in the documents “Maintenance_LI7700”, “Maintenance_LI7200RS” and “Maintenance_biomet_sensors” accordingly.

Calibrations

- Calibrate the gas analyzers LI-7700 and LI-7200RS ideally in the beginning of every season, following the details explained in the “Maintenance_LI7700” and “Maintenance_LI7200RS” documents.

Fieldwork Protocol & Procedures

Planning a trip

Before any visit make a clear list of tasks and goals. If you are going to one of the one of the flux stations you can check the field logs. See what was done on the last visit and check for notes to see if there is anything that needs to be done.

Scheduling and communication

Let the rest of the team know when you will be going into the field and make sure you have at least one person joining you.

Vehicles options (UQAM)

We have a few options to get a car to go to the field:

- Ask Louis-Philippe Toupin for a car that meets your requirements.
- Book a car from the GRIL lab.

Safety

Ensure you are familiar with the “Safety_Protocols” document.

In Summer:

- Be caution of fire safety.
- Bring extra water to drink.
- Bring proper clothing to protect yourself from the sun and nature.
- Be mindful of weather forecasts and avoid visits during storms.
- Bring the necessary PPE to ensure a safe visit.

In Winter:

- Be mindful of the weather forecasts
- Dress appropriately, bring extra socks, a head lamp, gloves, water, food, etc.
- Bring the necessary PPE to ensure a safe visit.

Daily Site Monitoring

Below are instructions for visual checks of the data that should be done every day.

Check Critical Variables

Motor Power (%) & Flow Fate for LI-7200 pump

- Keep checking if this one changes over time (over weeks).
    - Flow drive % should be 50 - 90%.
    - Higher power means more air-flow resistance in the tubing indicating either clogged up filter, dirty rain cap wire-mesh or damaged tubing.
- We aim for 30-min averages flow rate of slpm = 15 L/min.
    - Below 13 L/min we begin to worry, (data quality drops).
    - Below 10 L/min we need to fix it ASAP (data quality questionable or data missing) by cleaning up the intake tube cap.

Thermocouples

- LI-7200: these two values should be very close to each other (<1 C difference)
    - In (t_in_LI_7200)
    - Out (t_out_LI_7200)
- LI-7700:
    - Air temperature

Biomet Data

- Check if there are new data and it the values are reasonable for all sites.

Flux Data

For **Fluxes and Gas Concentrations** check if there are new data & if the values are reasonable. These will only be calculated if smart flux is running properly.

Gas Concentrations:

- CO2 mixing ratio (dry) - during daytime should be around 380-420 ppm (umol/mol), during nighttime it can be quite high depending on the site (sometimes > 600 ppm).
- CH4 mixing ratio (dry) - should be around 2 ppm

Fluxes:

- Sensible & Latent heat - Typically between -100 and + 450 w m^{-2}
    - Should generally sum ~ 70%-80% of net radiation.
- FCO2 & FCH4 - site dependent, but look for gaps, extreme spikes, systematic jumps.

Signal strength

- View over a longer period to check for any trends. Schedule a site visit if it's decreasing rapidly (<10%). It's normal for 7700's signal strength to drop during rainy periods, but it's not normal for 7200's to drop that frequently (if it does, check the 7200 head's o-ring).

Voltage

- Look at data over the past month and David or Zoran if the battery voltage goes below 24 V at midnight.
    - This usually happens in the winter when we have a few days of clouds & rain.

Remote Login Procedures

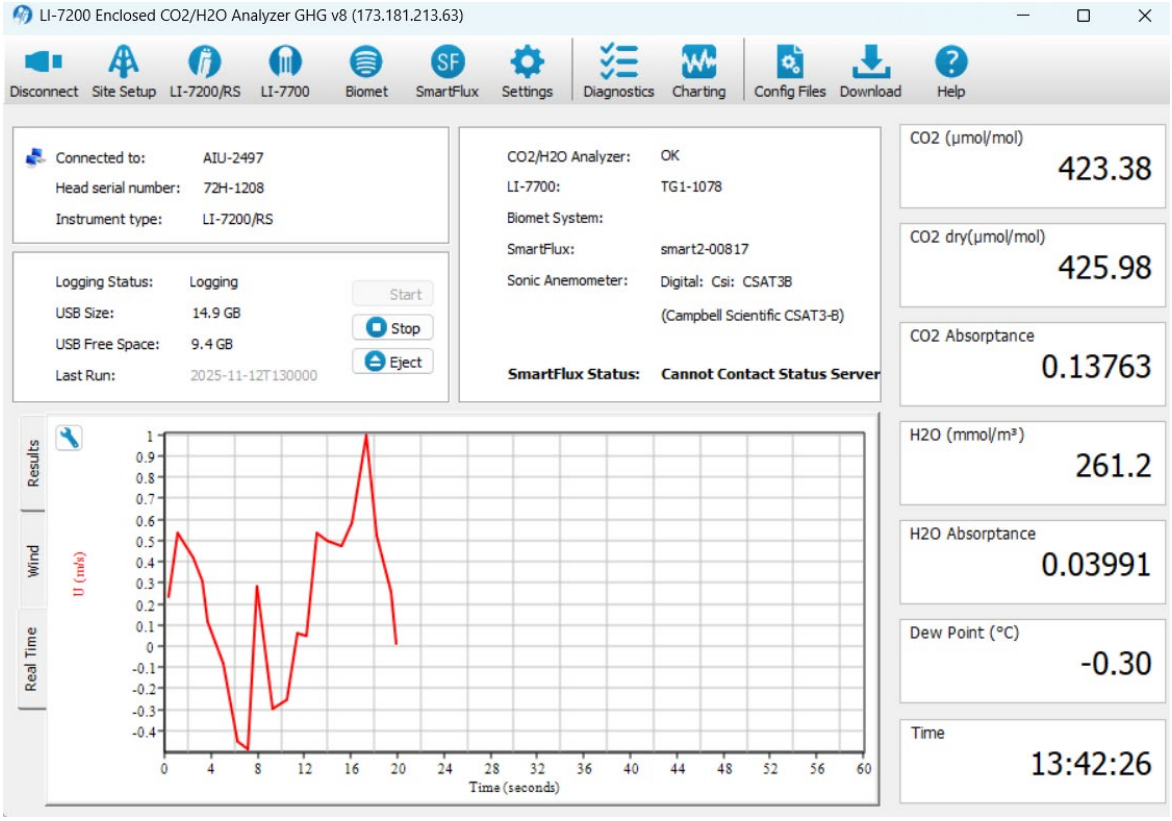
Use the LI7xx0 software to connect to the desired station.

The fixed IP addresses of each station are:

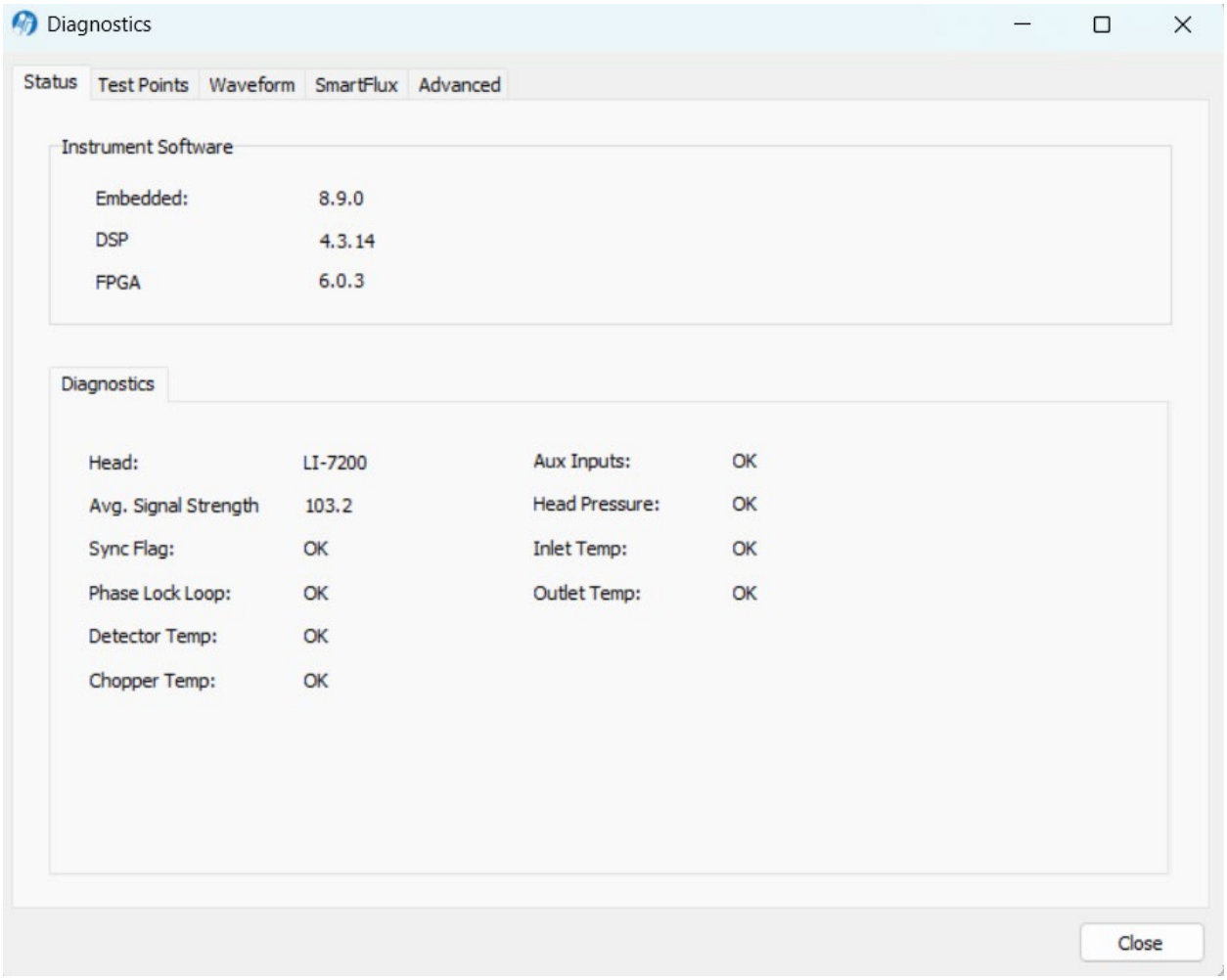
- UQAM_1: 68.182.132.135
- UQAM_2: 68.182.132.133
- UQAM_3: 96.1.34.212
- McGill_1: 173.181.213.63

Once on the main page check:

- The logger status → should be logging
- USB Free Space → make sure there's still enough space in the USB (usually the USB stick can handle 6-weeks data)
- The SmartFlux module must be connected to the system, make sure it's not "none"
- Head pressure - should be between -0.8 to -3.8 kPa
- Flow drive - should be around 50% to 90%
- Flow rate - should be around 13-15 l/m



- Click on the “Diagnostics” tab to check if all parameters are “OK”.



To check the LI-7700 connect to the LI-7700 software using the same fixed IP but with the port 7700.

- Check the 7700 Signal Strength, if it is 30-40% plan a visit to clean the sensor. Ideally the sensor would never be under 20%. Using the manual setting will allow you to apply some fluid and spin the mirror, this could do the trick but probably won't help much since the cleaning it is already automated throughout the day.
- Check 7700 Optics RH. If it's ≥ 30%, we should replace the internal chemicals.

Check SmartFlux internal memory every 2-3 months